What a resting echocardiogram cannot tell you

Identifiability of hidden tissue parameters in a mechanistic model of hypertrophic cardiomyopathy under myosin inhibition

Abstract

Two patients with hypertrophic cardiomyopathy can share an ejection fraction the day before myosin-inhibitor treatment begins and diverge completely under the same dose escalation: roughly one in twenty crosses the 50% ejection-fraction floor that triggers interruption. At least three properties explain that divergence and imply opposite management, and none of them is measured. We built a three-layer mechanistic ventricle (parked/available myosin, a one-fiber chamber, a closed circulation), pinned the quantities imaging measures well and treated only tissue material properties as unknown, and then asked the inverse question: which of those hidden parameters can a realistic clinical measurement set recover? The model reproduces 35 of 35 prespecified validation gates. It predicts a 5.3-point fall in ejection fraction at the mid dose against a placebo-corrected 4.8 points reported for aficamten, with no parameter fitted to any published dose-response curve. It under-predicts the rate at which patients cross the ejection-fraction floor (1.4% against a published 5.2%), which we report rather than tune away. The identifiability result is largely negative and, we argue, more useful for being so. **Drug clearance explains 50% of the variance in who over-responds and is *structurally* invisible before the first dose**: it enters the model only through drug exposure, so every derivative with respect to it is exactly zero and no provocation maneuver can recover it. Of the four remaining parameters, 2 are recoverable from a routine study at realistic measurement error. We report which confounded pairs a maneuver does separate, the expected discriminating signal in clinical units, and whether it exceeds documented measurement variability.

1 The question

Picture two patients on the day before treatment starts. Same symptoms, same obstruction, same wall thickness, *same ejection fraction.* On the measurement that governs their safety they are one data point, not two. Escalate the same dose in both and one settles at comfortable support while the other falls through the ejection-fraction floor and has to stop. In the long-term extension of the pivotal mavacamten programme this happened to 5.2% of patients at interim analysis and 8.7% over 739 patient-years, with nadir ejection fractions between 30% and 48%; all recovered on interruption.

Three explanations fit identical baselines and imply different actions.

- **Less contractile reserve.** Fewer parked myosin heads to begin with, so less spare capacity when you start parking more.
- **Stiffer tissue.** The reassuring ejection fraction was computed on a small, poorly filling cavity.
- **Slow clearance.** Twice the exposure from the same pill. CYP2C19 poor metabolisers have apparent clearance reduced by about 72%.

The first two mean the heart is fragile. The third means lower the dose and carry on. Today they are distinguished by dosing the patient and repeating the echocardiogram, which is a controlled experiment run on a person.

Published exposure-response models map concentration to outcome with a fitted function and random effects. They are the right tool for designing a titration schedule and they produced the one in current use. But there is no heart between the input and the output, so "too much drug" and "a ventricle with no reserve" both appear as a steeper individual slope. Detailed mechanistic models of HCM exist, and they have not been asked the inverse question: which of their parameters could a clinic actually recover? **That question decides whether a mechanistic model of HCM can ever be personalised from clinical data, as opposed to being illustrative.**

What we did, in four steps. Stated plainly, because the rest of the paper is the execution of exactly this:

1. **Build a ventricle whose knobs are the biology.** A three-layer mechanistic model whose inputs are tissue and molecular quantities (how many myosin heads are available, how calcium-sensitive the filaments are, how stiff the passive tissue is, how fast the drug is cleared) and whose outputs are the things a clinic measures.
2. **Find which of those knobs decide who over-responds.** Sample 5005 virtual patients, dose each one, and compute total-order Sobol indices of every hidden parameter on the ejection-fraction drop. This answers "what actually drives the risk".
3. **Find which of those knobs a clinical study can see.** Two levels: sensitivity of each measurement to each parameter, and then the harder question of whether the joint map from measurements back to parameters can be *inverted* at all, via Fisher information and posterior sampling under a documented noise model.
4. **Ask whether a stress test rescues what is invisible.** Search four provocation maneuvers for one that narrows a confounded direction by more than measurement error.

Steps 2 and 3 produce two rankings of the same five parameters, and comparing them is the result: see Section 5.2.

What this work produces. Four things, all in the repository and all regenerated by one command:

1. A **validated mechanistic ventricle** that reproduces 35 of 35 prespecified physiological gates and, unprompted, the HCM phenotype.
2. A **sensitivity matrix** (Figure 3) saying which hidden tissue parameter each clinical measurement can see, and by how much.
3. A **confounding map and recovery table** (Figure 4, Table 2) saying which parameters a realistic echocardiogram recovers and which it cannot, at two documented levels of measurement error.
4. A **tie-breaker table** (Table 3) saying, per confounded pair, whether any provocation maneuver resolves it and whether the discriminating signal exceeds documented measurement variability.

Every figure ships with the CSV behind it. The conclusions those four artifacts support, and the ones they do not, are set out in Section 6.

2 The model

Three layers, coupled, zero-dimensional throughout. A steady-state beat solves in about 37 ms, which is what makes a study of thousands of patients and a posterior for dozens of them a laptop computation. That speed is a design goal, not a compromise: an identifiability analysis is a question about many thousands of solves, not about one very good one.

Layer 1: myosin. A three-state population, parked (S), available (D) and attached (A), with $S + D + A = 1$. The resting equilibrium is one interpretable number,

$$\phi = \frac{k_{\text{park,off}}}{k_{\text{park,off}} + k_{\text{park,on}}},$$

the fraction of unattached heads that are unparked. **ϕ is what "HCM" means in this model**: healthy is low (0.35), HCM raises it (0.55), and the drug lowers it. Both rate constants are derived from ϕ and a fixed total turnover, so disease and treatment move one quantity in opposite directions. Thin-filament activation is a Hill function of calcium with a length-dependent Ca_{50}, and ATP is charged one per attachment event, so parked heads cost nothing.

The specification this work followed prescribed $\sigma_a = T_{\text{ref}} A$, and three additions were needed, each because the model visibly failed without it. An **active force–length** term, because cavity pressure is $\sigma_f/(1 + 3V/V_w)$ and so at fixed fiber stress a *smaller* cavity makes *more* pressure: with no length dependence the chamber's pressure-generating capacity rises as it empties (141 mmHg at 100 mL, 224 at 42, 314 at 13), a *negative* end-systolic elastance, and ejection never meets a force balance. **Cross-bridge distortion**, giving a force–velocity relation, because without it shortening speed is unbounded and the first coupled run ejected past zero volume. And **force-dependent recruitment from the parked state**, the published mechanism for length-dependent activation, because calcium-sensitivity length dependence alone gave a stroke volume that was flat then falling as preload rose; it is also what makes contractile reserve mechanistically real, since a depleted parked pool cannot be recruited when load rises.

Layer 2: chamber. The one-fiber relation of Arts and colleagues: $\varepsilon_f = \frac{1}{3}\log((V_{\mathrm{lv}}+V_w/3)/(V_{\mathrm{lv,ref}}+V_w/3))$ and $p_{\mathrm{lv}} = \sigma_f/(1+3V_{\mathrm{lv}}/V_w)$. **This layer carries the thesis.** V_w and V_{lv} are both measured well clinically, so they are pinned per virtual patient and never inferred; only Layer 1's material parameters and the drug clearance are hidden. Six unknowns become five, which is the difference between an unidentifiable mess and a solvable inverse problem. The split is enforced as a type distinction in the code, not a convention.

Layer 3: circulation. A closed loop conserving stressed volume by construction. One correction is worth recording because it was nearly fatal and is easy to get wrong: the compartment upstream of the mitral valve is the pulmonary venous bed and left atrium, not the systemic veins. At a systemic venous compliance the reservoir absorbs whatever the ventricle declines to accept, and end-diastolic pressure becomes *insensitive to the ventricle's own stiffness*. Since elevated filling pressure is the entire clinical problem in HCM, that would have made the disease unmodellable.

Obstruction and drug. Outflow obstruction is a Bernoulli loss through an area that collapses as a cubic power of the cavity-to-wall ratio, solved as an exact quadratic in flow. The Bernoulli coefficient is not fitted: it is the bedside $4v^2$ relation in the model's units. The drug maps a maintained dose to a steady-state concentration through a per-patient clearance and thence to a saturating reduction in ϕ.

3 Validation

35 of 35 gates pass (all: yes). The healthy reference returns an ejection fraction of 0.658, an end-diastolic volume of 113.0 mL, a stroke volume of 74.4 mL, a peak left-ventricular pressure of 110.8 mmHg, a filling pressure of 9.6 mmHg, a mean arterial pressure of 94.9 mmHg, a cardiac output of 4.85 L/min and a wall thickness of 0.92 cm. Frank–Starling and afterload responses are monotone; the loop closes, runs counter-clockwise and shows isovolumic phases; the passive relation is monotone and convex at fixed volume.

Three gates deserve comment.

The disease emerges. Raising availability and stiffness with a thicker wall yields a supranormal ejection fraction (0.747 against 0.658), a reduced stroke volume, an elevated filling pressure, a raised filling-pressure surrogate, reduced strain despite preserved ejection fraction, and roughly twice the ATP cost per unit of external work (Figure 1). None of these is an input.

Wall thickening alone raises ejection fraction. Giving the HCM geometry *healthy* material returns an ejection fraction of 0.758, slightly *above* the diseased reference. This is correct physiology, and it is the sharpest available statement of why the project exists: **in a thick-walled ventricle a reassuring ejection fraction is close to uninformative**. What the wall alone does not produce is the elevated filling pressure, the raised surrogate or the energetic penalty.

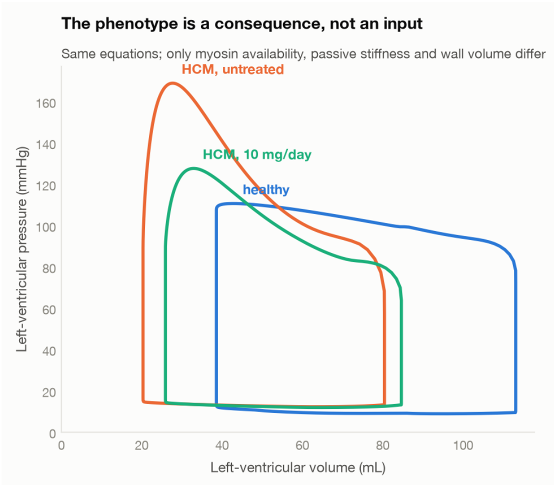

Figure 1: The phenotype is a consequence, not an input. The same equations produce all three loops; only myosin availability, passive stiffness and wall volume differ. Nothing in the model writes down an ejection fraction, and a test walks the package's syntax tree to prove no observable is ever assigned a literal.

An unfitted exposure-response prediction. The reference patient loses 5.3 ejection-fraction points at the 10 mg/day mid dose while its outflow gradient falls by 39.4 mmHg. SEQUOIA-HCM reported a placebo-corrected change of −4.8 points against a gradient falling from about 55 to about 20 mmHg. **No parameter in the model was calibrated against any published dose-response curve**, which the provenance table documents constant by constant, so this is a prediction rather than a fit.

Calibration honesty. References were calibrated against resting haemodynamic *ranges*, never a trial outcome. Three ejection-realism targets were added after a first calibration passed every haemodynamic gate while ejecting the whole stroke volume in 154 ms at twice physiological peak flow: a model can satisfy its checks and still be wrong.

4 Virtual population

5005 patients were drawn on a Saltelli design over eleven inputs: three measured geometric, five hidden material, three loading. Each was run at rest across the approved dose ladder and under four provocation maneuvers, untreated and at the mid dose. **Applying the pivotal trial's own enrolment criteria** (ejection fraction $\geq 55\%$, hypertrophy on imaging, gradient ≥ 30 mmHg at rest or 50 provoked) leaves 1834 patients with a median ejection fraction of 0.717, a median gradient of 67 mmHg and a median wall thickness of 1.64 cm.

1.4% of them cross the ejection-fraction floor at or below the mid dose, against a published 5.2% over a full titration. **The model under-predicts this rate and we report the discrepancy rather than closing it.** Three reasons are visible in the cohort. The eligible virtual patients are more severely obstructive than the trial's (99% obstructive, median gradient 67 mmHg against roughly 55) and start from a higher ejection fraction, so they have further to fall. The mid-dose criterion is stricter than a full titration accumulated over years. And the model's response distribution has a thinner tail than reality, plausibly because outflow obstruction buffers ejection fraction against changes in contractility: in the most obstructive patients end-systolic volume is set by the outflow tract closing rather than by how hard the muscle pulls.

Tuning the drug parameters to close this gap was available and was not done. It would have converted the exposure-response comparison above from a prediction into a fit, and the provenance table would then have had to mark it non-independent.

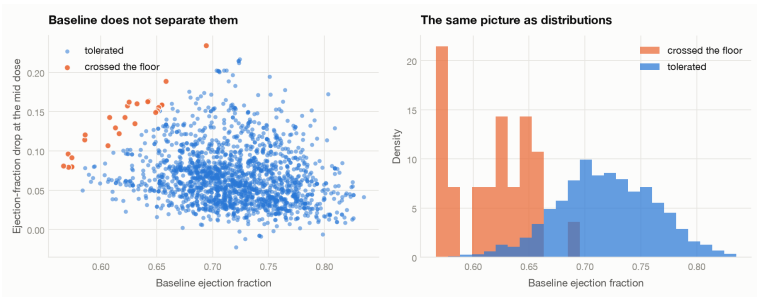

Figure 2: The premise, drawn. Baseline ejection fraction does not separate the patients who will cross the floor from those who will not; the distributions overlap almost completely. This is the problem the rest of the paper is about.

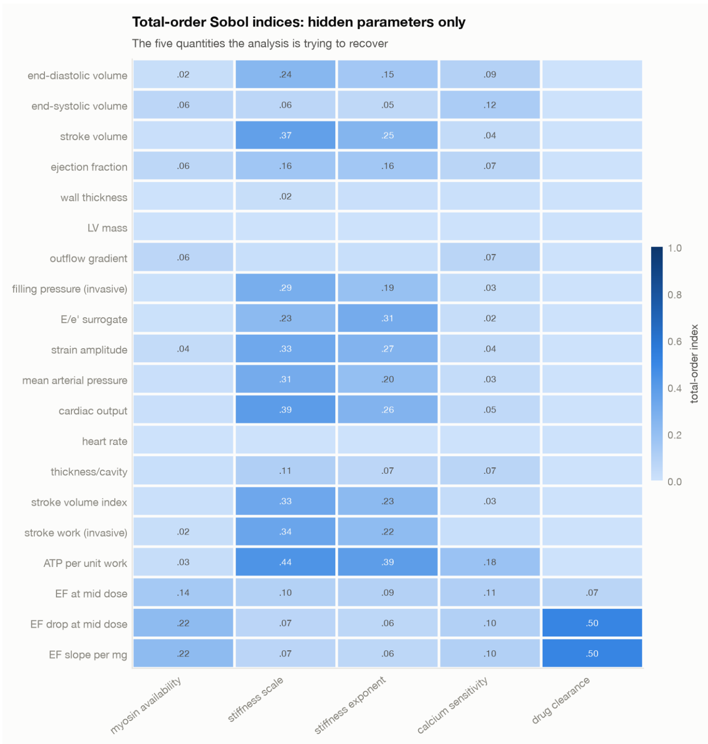

Figure 3: Total-order Sobol indices of each observable and outcome against the five hidden parameters. The bottom rows are outcomes, not measurements. Note the clearance column: everything above the outcome rows is zero.

5 What the measurements can and cannot see

5.1 Sensitivity

Total-order Sobol indices were computed for every observable and every sampled parameter (Figure 3). Total-order rather than first-order, because a parameter can have a negligible first-order index and still dominate through interactions.

The single most consequential finding is visible immediately. **drug clearance explains 0.50 of the variance in the ejection-fraction drop at the mid dose, and its total-order index on every baseline observable is 0.000.** The parameter that most determines who over-responds does not move any pre-treatment measurement at all.

5.2 The central result: importance and visibility run backwards

Step 2 asked which hidden parameters drive over-response. Step 3 asked which ones a measurement can see. Table 1 puts the two rankings side by side, and they run in opposite directions.

Hidden parameter	Drives outcome	Rank	Best visibility	Rank	Recovered?
drug clearance	0.505	1	0.000	5	no
myosin availability	0.220	2	0.063	4	yes
calcium sensitivity	0.100	3	0.181	3	yes
passive stiffness scale	0.067	4	0.436	1	no
passive stiffness exponent	0.060	5	0.389	2	no

Table 1: Left: how much each hidden parameter drives the ejection-fraction drop (total-order Sobol index). Right: the largest total-order index that parameter achieves on *any* observable, which is the best case for measuring it. The two rankings are reversed, with only the bottom two swapped. Generated from `results/visibility_vs_importance.csv`.

The parameters a clinical study sees best are the ones that matter least for the outcome, and the parameter carrying half the outcome variance is the one it cannot see at all. The Spearman rank correlation between driving the outcome and being visible is -0.90 ($p = 0.037$), and it is identical if the analysis is restricted to routine non-invasive observables only. Reading the same table by variance rather than by rank: the two parameters a realistic study recovers account for 0.32 of the total-order index on the outcome, while the three it misses account for 0.63. The clinic gets the third of the risk it can see, and loses the two thirds it cannot.

Five parameters is a small sample, so read the sign and not the decimal. A rank correlation over five items is a descriptive statement about this model's parameter set, not an estimated quantity with a meaningful confidence interval. The individual entries are the evidence; the correlation is a way of summarising them. What is *not* sample-dependent is the extreme entry: clearance is at zero visibility for a structural reason given in Section 6, not because five was too few.

Why the two steps of "can we see it" are both needed. Visibility in Table 1 is necessary and not sufficient, and the table contains an example in each direction. Myosin availability shifts no single observable much (0.063) and is nonetheless recovered, because several weak signals combine into one well-determined direction. The two passive stiffness parameters are the most visible in the whole model (0.436 and 0.389) and neither is recovered, because they are visible through the same combination: the map is many-to-one, and high sensitivity buys nothing against it. This is why Section 5.2 does not end the paper and the posterior analysis follows.

5.3 Identifiability

Local. Fisher information matrices were built by finite-differencing the observable vector with respect to the logarithm of each hidden parameter at 50 representative operating points, weighted by measurement error. The smallest eigenvalue is 0.000000 against a largest of 188.5: **the matrix is exactly singular**, and its null direction is 100.0% clearance.

This is not a numerical curiosity. Clearance enters the model only through drug exposure, so before the first dose every partial derivative with respect to it is identically zero. The distinction that matters clinically is between a parameter that is *weakly identified* (the measurements respond, but not enough to beat the noise; a better maneuver or machine can help) and one that is *structurally unidentified* (the measurements do not respond at all; no maneuver can amplify a signal that does not exist). **Clearance is the second kind. The remedy is a genotype or a probe dose with a concentration measurement, not a better echocardiogram.**

Global. For 50 patients the observables plus realistic noise were treated as data and the posterior recovered by Markov chain Monte Carlo, against a per-patient polynomial emulator fitted on 400 design points (held-out R^2 at worst 0.6547, typically 0.99902). The emulator's error is added in quadrature to the measurement noise so imprecision widens the posterior rather than corrupting it, and the synthetic data always come from the exact model so emulator error acts as misspecification rather than cancelling out.

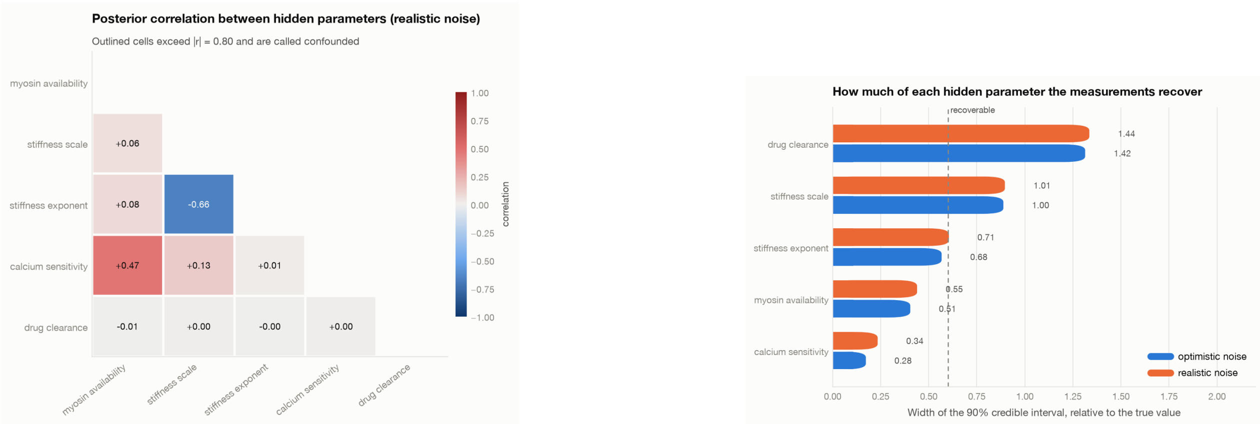

Figure 4: Left: posterior correlation between hidden parameters at realistic measurement error. Right: how much of each parameter is actually recovered. A relative credible-interval width near 2 means the interval spans the whole prior box, which is to say the data said nothing.

At realistic noise the most strongly coupled pair is passive stiffness scale and passive stiffness exponent at $|r| = 0.68$; at optimistic noise it is passive stiffness scale and passive stiffness exponent at 0.81. Of the five hidden parameters, 2 are recoverable (calcium sensitivity, myosin availability), and the rest are not (passive stiffness exponent, passive stiffness scale, drug clearance). The full table is Table 2.

Noise level	Hidden parameter	Rel. 90 pct CI width	Median abs. bias	Recoverable?
optimistic	ca50 ref um	0.28	0.05	True
optimistic	phi baseline	0.51	0.09	True
optimistic	b pas	0.68	0.13	False
optimistic	a pas kpa	1.00	0.24	False
optimistic	clearance l per h	1.42	0.34	False
realistic	ca50 ref um	0.34	0.06	True
realistic	phi baseline	0.55	0.09	True
realistic	b pas	0.71	0.16	False
realistic	a pas kpa	1.01	0.22	False
realistic	clearance l per h	1.44	0.35	False

Table 2: Parameter recovery at both noise levels. "Recoverable" means the median relative 90% credible interval is below 0.60.

5.4 Tie-breaker search

For each confounded pair and each candidate maneuver we recomputed the posterior using baseline observables *plus* the maneuver's. The signal is computed by displacing the truth along the *confounded direction* by the amount a resting study cannot resolve, and asking how far apart the two indistinguishable patients look under the maneuver.

A note on the scoring criterion, because the obvious one is wrong. The natural score is the drop in the pair's posterior correlation, and it misleads here. **Correlation describes the shape of the uncertainty, not its size.** Adding information that constrains the same combination the data already knew collapses the posterior further onto its ridge, so the correlation goes *up* while the patient becomes better characterised. That is what happens for two of the three pairs examined: correlations rise from 0.47 to 0.72 and higher under every maneuver. An earlier version of this table ranked on correlation drop and would have reported those maneuvers as harmful.
The table therefore ranks on how much the maneuver narrows the posterior *along the direction that was invisible at baseline*: the posterior standard deviation projected onto that fixed direction, before and after. That is the question actually being asked.

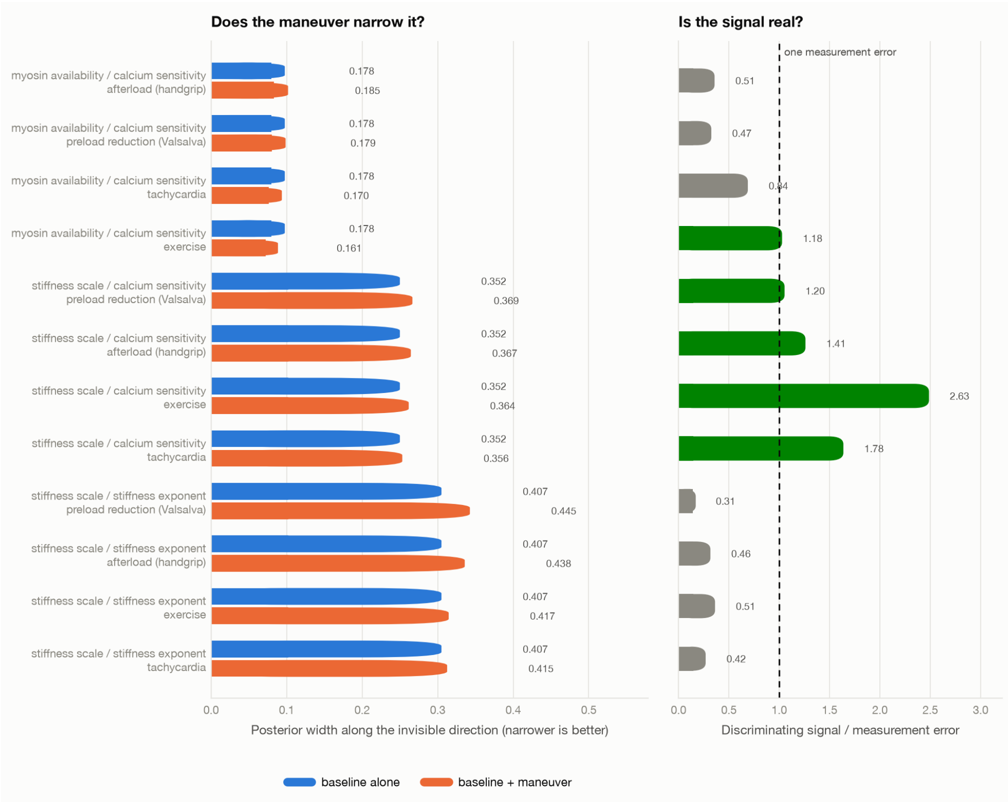

Figure 5: Does the maneuver break the confound (left), and is the resulting signal bigger than the error bar (right)? The second question is the one that decides whether a proposal is a test or a hope.

Confounded pair	Maneuver	Narrowed	Discriminating	Signal	S/N	Beats noise?
availability / Ca sens.	exercise	0.054	strain amplitude	0.0049	2.19	True
stiff. scale / Ca sens.	exercise	0.046	strain amplitude	0.0117	8.63	True
stiff. scale / stiff. exponent	tachycardia	-0.021	strain amplitude	0.004	0.96	False

Table 3: The tie-breaker table. Ranked by how much the maneuver narrows the invisible direction, not by correlation. "Beats noise?" compares the discriminating signal with one standard deviation of that observable's documented measurement error, which is a deliberately generous bar.

The best case is myosin availability and calcium sensitivity under combined exercise, where the width along the invisible direction falls from 0.177 to 0.161, a 5.4% reduction. 2 of 3 examined pairs produce a discriminating signal above one measurement error, but only 1 show a meaningful narrowing of the invisible direction. The maneuver was interpretable in 100% of the patients it was applied to; any remainder are ventricles too stiff to fill at the provoked heart rate, which is not a numerical failure but the maneuver being uninterpretable in exactly the patients it was proposed for.

The two columns disagree, and the disagreement is the result. The exercise maneuver moves strain amplitude by 0.49 percentage points between two patients a resting study cannot tell apart, which is 2.19 measurement standard deviations and looks decisive. Yet the posterior barely narrows along the direction that signal was supposed to resolve. Three candidate explanations, two of which we can rule out.

It is not the emulator: its held-out error is at most a quarter of the measurement error for any observable in any condition, so adding it in quadrature inflates the likelihood width by about 3%. It is not sampling noise: the pattern is consistent across all fifty patients and all four maneuvers.

It is nuisance compensation. The signal is computed with the other three hidden parameters held at their true values, and a real inference is not granted that. A change in the maneuver's observables that is large when the rest of the tissue is known can be mimicked by adjusting the parameters that are not, and an inference estimating all five at once cannot separate the two explanations. So the honest reading is that these maneuvers *do* move the measurements, and that the movement is not specific enough to the confounded combination to resolve it once everything else is also unknown. That distinction matters for anyone designing the study this analysis is meant to inform. A protocol justified by "the maneuver changes the measurement a lot" is justified by the wrong quantity.

6 Conclusions

In one sentence. **The tissue and molecular parameters that decide who over-responds to myosin inhibition are, in this model, close to the complement of the ones a clinical study can recover: rank correlation -0.90 between mattering and being visible, with 0.63 of the outcome-driving index sitting on parameters a realistic echocardiogram does not recover.** Everything below is that sentence taken apart.
Stated as separate claims, each with how much weight it will bear. The distinction matters: one of these follows from the structure of the problem and would survive a different model, and the others are properties of *this* model under *these* error bars.

1. Drug clearance is structurally unidentifiable before the first dose. **Confidence: high, and independent of our calibration.** Clearance enters the model only through drug exposure, so at zero dose every partial derivative with respect to it is identically zero, the Fisher information matrix is exactly singular, and its null direction is 100.0% clearance. This is arithmetic, not a numerical result: it would hold for any model in which clearance acts only through exposure, which is to say for any pharmacologically sensible one. Since clearance also carries half the variance in who crosses the floor (total-order Sobol index 0.50), **the single largest determinant of over-response cannot be recovered from any pre-treatment measurement, however good.**

2. No provocation maneuver rescues it, and none of them resolves the tissue confounds either. **Confidence: medium.** For clearance the failure is structural, so the conclusion is as strong as claim 1: a maneuver amplifies signal, and there is no signal. For the tissue parameters the finding is empirical and therefore weaker: across four maneuvers and three confounded pairs, the best narrowing of an invisible direction was 5.4%. The direction of that result is safer than its magnitude.

3. Two of the five hidden parameters are recoverable from a routine study; three are not. **Confidence: medium; depends on the noise model.** Recoverable: calcium sensitivity, myosin availability. Not: passive stiffness exponent, passive stiffness scale, drug clearance. The two passive stiffness parameters trade off against each other, which is unsurprising for two arms of one exponential, and their confound is the one that a better measurement rather than a better maneuver would address.

4. A high ejection fraction in a thick-walled ventricle is close to uninformative. **Confidence: medium; a model result, but a mechanistically transparent one.** Wall thickening alone, with entirely healthy tissue, raises ejection fraction to 0.758, above our diseased reference. What the thick wall does *not* produce is the raised filling pressure, the raised E/e' surrogate or the energetic penalty. Those need the material change, and they are what make a patient symptomatic.

5. Posterior correlation is the wrong way to score a proposed maneuver. **Confidence: high; a methodological point that generalises.** Correlation describes the shape of the uncertainty and not its size, so information that constrains the combination the data already knew raises it while improving the inference. Score the width along the direction that was invisible instead. Relatedly, a discriminating signal computed with the other parameters held at truth is an upper bound and can be several-fold optimistic, because a real inference can mimic the maneuver's effect by moving what it does not know.

What we cannot conclude. That any of this describes a particular patient: nothing here is validated against one, and the comparisons to trial data are aggregate-to-aggregate. That the absolute numbers are right: a third of the model's constants are labelled `assumed`. That the tie-breaker result would survive a richer model: a maneuver that fails here might succeed in a model with regional mechanics or catecholamine-dependent relaxation, both of which this one lacks.

A caution about the direction of our errors. **Several modelling choices make the posterior tighter than reality.** The noise model treats observables as independent when their errors are correlated; it is Gaussian where real error has heavier tails; and the outflow-gradient noise used (35%) is deliberately below the 46–52% coefficient of variation the literature reports for that quantity. So "these parameters remain confounded" is a safe conclusion here and "this maneuver separates them" is a fragile one.

7 Where to look next

7.1 Clinically

Get the pharmacokinetics before titrating, not from it. This is the direct implication of claim 1, and it is actionable now. If clearance is both the dominant driver of over-response and invisible to imaging, then the information has to come from somewhere else: a CYP2C19 genotype, or a probe dose with a single concentration measurement before escalation begins. Current clinical interest in prospective genotyping is pointed the right way; this work supplies a mechanistic argument for why echocardiographic titration alone cannot substitute, rather than an empirical observation that it did not.

A testable prediction, checkable in data that already exists. Patients matched on baseline echocardiography but differing in measured exposure should show systematically different dose-response slopes, and patients matched on *exposure* should show much less residual spread than patients matched on echocardiography. Both are post-hoc analyses of trial datasets that already contain paired pharmacokinetic and echocardiographic measurements. If the second comparison shows a large residual spread, claim 1 is incomplete and something in the heart is carrying more of the variance than this model gives it.

For the tissue parameters, measure them rather than provoke them. The passive stiffness confound is not a maneuver problem. A maneuver moves the operating point along a relation the study already cannot resolve; what would separate the two arms of the exponential is sampling the relation at more than one point, which means a direct stiffness measurement (shear-wave elastography, or invasive pressure-volume at two loading states) rather than a harder stress test.

Do not design a protocol on effect size alone. Claim 5, in clinical dress. A maneuver that visibly changes a measurement is not thereby diagnostic, if the change can be produced by something else you also have not measured.

7.2 Computationally

In rough order of how much each would change the conclusions:

1. **Growth and remodelling**, so hypertrophy develops over simulated months rather than being sampled. This is the largest structural gap: it would let the model say how a geometry-plus-material combination arose, and whether the measured/hidden split is stable over a disease course.

2. **Regional mechanics**, via a three-dimensional idealised ventricle, to test whether the zero-dimensional conclusions hold once asymmetric hypertrophy, fibre disarray and systolic anterior motion are representable. Claim 2 is the one most likely to move.
3. **Differentiable inversion** (a JAX reimplementation), which would replace the per-patient polynomial emulator with exact gradients and make the identifiability analysis cheap enough to run over many more patients and maneuvers.
4. **Geometry personalisation** against a public cardiac-MRI cohort, which is what would turn "pinned geometry" from a modelling convenience into a measurement.
5. **Catecholamine-dependent relaxation**, which the model lacks and which is the most likely reason its filling pressure fails to improve on treatment where trials report that it does.

Three measurements the literature owes this model. Recorded as `[GAP]` in the dossier rather than filled with plausible values: the parked-versus-available myosin fraction in healthy and HCM *human* myocardium; a representative-fibre passive stiffness pair fitted to human tissue under this constitutive form; and test-retest reproducibility for maximal wall thickness in an HCM cohort. The first two are why two of the model's most important constants are labelled `calibrated` rather than `measured`.

8 Limitations

Stated plainly, because several of them bound the conclusions rather than merely qualifying them.

- **Prescribed calcium, not simulated electrophysiology.** No ion channels, no rate-dependent calcium handling, no catecholamine effect. The tachycardia and exercise maneuvers therefore capture the shortening of filling time and nothing else, which is the mechanically dominant effect in a stiff ventricle but is not the whole of exercise. The calcium time constant was calibrated to systolic *duration* rather than to the measured upstroke, because a single-time-constant waveform cannot match both.
- **Zero-dimensional chamber.** Homogeneous wall loading: no fiber disarray, no regional fibrosis, no asymmetric hypertrophy (which is what HCM characteristically has), and no systolic anterior motion as an explicit mechanism.
- **Steady-state pharmacokinetics only.** The model can say *whether* a maintained dose takes a patient below the floor and not *when*. The clinical protocol is fundamentally about timing, so this bounds what can be claimed.
- **No growth or remodelling.** Wall volume is sampled, not grown, so the model says nothing about how a given geometry arose or where it is heading.
- **The drug slightly raises filling pressure in the model,** whereas trials report improved filling parameters on treatment. The model has no load-dependent improvement in relaxation. This is a real discrepancy and is reported rather than smoothed over.
- **A third of the model's constants are labelled `assumed`** in the provenance table (10 measured, 22 calibrated, 33 assumed). Absolute outputs should be read as illustrative; directional and relative outputs are the substantive ones.
- **No mutation-specific claims, and nothing validated against a patient.** The mapping from a named variant to a numerical parameter is literature-derived and approximate, so every claim here is about regions of parameter space; the comparisons to trial data compare aggregate behaviour to published aggregates.

9 Reproducibility and transparency

One command reproduces every figure, and every figure ships with the CSV behind it. `make all` inside the supplied container runs the test suite (all 35 validation gates plus a provenance-coverage test that fails if any constant lacks a sourced row), generates the cohort from a logged seed (20260816), and produces every table, figure and this document in 16.1 minutes. Every quantitative statement here is a macro resolved from a results file at build time; none is typed. The research dossier in `docs/research/` records the source of every constant, every simplification made relative to it, and every place the literature failed to supply what the model needed; those gaps are marked and described rather than filled with plausible values.

What was calibrated, and against what. The healthy and HCM reference patients were calibrated by grid search against resting haemodynamic *ranges* and three ejection-realism targets. **No parameter was calibrated against any published dose-response curve, any trial outcome, or the observed rate of ejection-fraction excursions.** The provenance table labels all 65 constants `measured` (10), `calibrated` (22) or `assumed` (33), so that claim is checkable rather than merely asserted. The two comparisons to trial data in Section 3 and Section 4 are therefore predictions; one of them agrees closely and the other does not, and both are reported.

Use of AI assistance. This project was implemented with **Claude Code** (Anthropic) working alongside the author. The assistant wrote the bulk of the Python implementation, the test suite, the figure code and the first drafts of the research dossier and this document, and carried out the literature search, the parameter calibration sweeps and the debugging of the model failures described in Section 2. Direction, scope, scientific judgement, and acceptance of every result remained with the author.

Two disclosures follow from that and are worth making explicitly, because they bear on how this work should be read. First, **the citations were checked, and the gaps were left open**: every reference in the bibliography was located and its content verified against the abstract or full text, and where a number the model needed could not be traced to a primary source the dossier says `[GAP]` and describes what would be required, rather than substituting a plausible value or a fabricated reference. Second, **several substantive findings emerged from failures rather than from design**: the negative end-systolic elastance, the filling pressure insensitive to its own stiffness, the misleading correlation metric. Each is documented in the source, the dossier and this paper together with the symptom that exposed it, so a reader can judge the fix rather than take it on trust.

Availability. Code, research dossier, results, figures and their underlying CSVs are in the repository, along with a pinned lockfile and a container image that reproduces the run described here from a clean checkout.